SMARTBRIDGE
Let's Bridge the Gap

Project Initialization and Planning Phase

Date	26 October 2026
Team ID	Gyanvi Jain
Project Title	Formula 1 Race Intelligence Dashboard
Maximum Marks	3 Marks

Project Proposal (Proposed Solution) template

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Project Overview	
Objective	To develop an interactive dashboard that simplifies Formula 1 data analysis by enabling users to compare driver performance, team standings, race reliability, tyre usage, and podium achievements.
Scope	The project covers data preparation, KPI calculation, interactive visualisation, driver and team comparisons, DNF analysis, tyre compound usage, geographic participation, and podium analysis using Tableau. The dashboard is published on Tableau Public, with optional Flask integration.
Problem Statement	
Description	Formula 1 data includes race results, driver statistics, team performance, circuit information, tyre usage, and championship standings. Analysing these datasets separately makes it time-consuming to compare performance across drivers, teams, and seasons and identify race failure patterns.
Impact	Users may spend additional time collecting and comparing information, making it harder to identify performance trends, understand DNF patterns, and communicate meaningful insights efficiently. An integrated dashboard can make these comparisons easier and support data-driven analysis.

Proposed Solution	
Approach	Collect Formula 1 data from Excel/CSV files, clean and validate the data, join relevant datasets, calculate performance metrics using Tableau calculated fields, and create interactive visualisations. Publish the completed dashboard on Tableau Public.
Key Features	1. Four KPI cards for teams, seasons, race outcomes, and circuits. 2. Driver performance leaderboard with a Top N parameter. 3. DNF reliability analysis by reason, driver, and team. 4. Team performance and championship comparisons. 5. Tyre compound usage analysis. 6. Geographic race participation visualisation. 7. Podium achievements by driver and team. 8. Interactive filters for DNF reason and race year.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	One laptop or desktop capable of running Tableau Desktop.
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	Approximately 5–10 GB of available storage is a practical allocation.
Software		
Frameworks	Python frameworks	Tableau Desktop and Tableau Public; Flask for optional web integration.
Libraries	Additional libraries	No additional analytical libraries are required for the Tableau-only implementation. HTML/CSS may be used for the optional web interface.

Development Environment	IDE, version control	Tableau Desktop, Tableau Public, and a web browser. Git may be used for version control.
Data		
Data	Source, size, format	Excel/CSV datasets comprising approximately 13,750 combined rows across five sheets, covering 17 seasons (2010–2026) and 29 circuits, as described in the project.